

Cross-machine wall-clock synchronization

NTP Server

Unix wall-clock

ESP32-P4

SNTP + time.exchange

t1 / t2

ESP32-S3

SNTP + time.exchange

t3 / t4

NTP-style estimate

$\text{offset} = ((t2 - t1) + (t3 - t4)) / 2$

$\text{delay} = (t4 - t1) - (t3 - t2)$

STM32 Node

mono tick + offset

STM32 Node

CTRL_TIME_SYNC 0x17

Deadlines and leases remain on monotonic clocks; wall-clock is only for cross-machine timestamps.